

BUILDING URBAN, BUILDING GREEN

The advantages of building a green home in a compact city center overrule the headaches of remediating a brownfield site for this Milwaukee family.

BY JULI KAUFMANN

In 2002, my husband, Mike Kaufmann, and I set out to better align our lives with our values of personal responsibility, commitment to urban community, and environmental sustainability. Since a house is the biggest financial investment and the most significant consumption of products we will likely make in our lifetime, we decided to build a green home as an embodiment of these values.

After extensive research, we chose a contaminated city lot located within the fabric of Walker's Point, a historic Milwaukee neighborhood. The city of Milwaukee had acquired the lot through foreclosure in 1980. At that time, older structures that had existed on the property had been demolished, leaving a vacant parcel of land. Because soil contamination was suspected, the lot sat vacant for nearly a quarter of a century, despite its location in a community with significant redevelopment potential.

An environmental investigative report, known as a Phase I Environmental Site Assessment, was finally commissioned by the city of Milwaukee in September 2000, in preparation for making the parcel available for redevelopment. This report identified the previous owners of, and uses at, the site, including a Pabst Guard Armory, a junk dealer, and a waste materials and paper facility. These historical uses validated the initial concerns about contamination at the site and prompted a Phase II Environmental Site Assessment, which includes testing of soil and groundwater at the site. These tests verified that contamina-



We chose to build a green home on a remediated site in a historic Milwaukee neighborhood.

tion was pervasive, distributed throughout the site, and, in some spots, elevated to hazardous levels. Since the previous owners either were unknown or could not be located, cleanup would be the responsibility of the current owner. This testing deterred any redevelopment at the site for a few more years.

Despite the known challenges, we pursued the site, beginning in 2002, as an ideal location for our home, since the site would allow us to restore, sustain, and enhance an existing parcel within an already-developed community. Besides providing us with the opportunity to clean up a brownfield site and return the vacant land to productive use, this lot was attractive for another reason as

well. It was an urban infill site—a site embedded within and already served by existing water, utilities, and transportation networks. Building on such a site helps to minimize the fragmentation of undeveloped land, and to reduce the associated environmental impacts of infrastructure extension. Compact development also promotes pedestrian activity and reduces vehicle miles traveled by encouraging walking, biking, or alternative means of transportation to necessary services. From our point of view, urban development encourages community connections and underscores our interdependence, a concept that is core to sustainability.

Preconstruction Hurdles

Following identification of the site, we undertook a two-year odyssey to acquire the lot from the Redevelopment Authority of the City of Milwaukee and to complete hazardous-waste cleanup in compliance with Wisconsin Department of Natural Resources brownfield remediation requirements. We quickly learned that building our ideal home was not going to be a simple undertaking. Very little about the project could

homes constructed in the United States. In addition, siting requirements pushed the home to the front of the lot and directly adjacent to a building to the south, effectively blocking passive-solar opportunities and limiting active-solar harvesting.

Bureaucracy. We had to learn how to navigate the overlapping and complicated requirements and approval process of the Redevelopment Authority of the city of Milwaukee, multiple city legisla-

ing relationships, an option not likely to be available to most families.

Brownfield Remediation

Despite these challenges, preconstruction work was finally completed and ground breaking occurred in November 2005. Before the structure itself could be built, the brownfield remediation had to be completed. Based on the results of the environmental testing of the site, we had only two options. The first option was to cap the entire site. This would entail making the site into a parking lot or covering the entire lot with a building that had no basement. Neither of these alternatives fit the plan to make the site suitable for a single-family home. Instead, we pursued the second option, which was to remove all surface level contaminants.

Approximately 6 feet of soil was excavated from the entire site and trucked away. Based on the testing results, most of the soil was categorized as special waste, meaning that while it was contaminated, it could be disposed of in a number of waste sites. One such waste site was the land beneath a major freeway reconstruction that was then underway in the Milwaukee area. We were able to truck the contaminated soil to this waste site and in doing so, assist with another local project. We were also able to use clean fill from the freeway construction site for soil on our site. Hence, the remediation actually created a win-win situation, and the only cost incurred was the cost of trucking the soil back and forth.

In addition to the special waste on most of the site, environmental testing also identified a few hot spots that contained hazardous levels of contamination. The soil in these hot spots had to be removed and treated at specialized landfills. Fortunately, a local landfill was an acceptable designated deposit point. The hazardous soil was treated and landfilled. This portion of the remediation included tipping fees—the fees charged by the landfill to accept the soil and cover the cost of treatment.



JULIE KAUFMANN

By eliminating most hallways, the Kaufmanns created a home that seemed larger than its square footage.

be handled like a typical home construction project, so almost everyone involved—including building contractors, lenders, insurers, civil servants, attorneys, architects, and so on—had to think outside of the normal home construction box. We faced a number of hurdles during this preconstruction phase before ground breaking could even occur.

Local zoning and building codes. Minimum frontage requirements mandated a home size that was larger than desirable by our standards, compromising the green objective to obtain a minimal footprint. We had originally hoped to construct a home under 1,500 ft². The completed project, while approximately 1,800 ft², is still smaller than most new

tive committees, the city historic preservation commission, the city Department of Neighborhood Services, and the Wisconsin Department of Natural Resources. As a result, the development process took more than two years, adding significant cost and complexity to the project.

Financing. Hazardous waste contamination on the site made it impossible to secure a clean “closure” ruling from the Wisconsin Department of Natural Resources. As a result, traditional financing could not be secured for the project. We had to negotiate an exceptional financing agreement, made possible by leveraging established bank-

Overall, the costs we incurred to remediate the site totaled approximately \$25,000 in excavation, trucking, back-filling, grading, and tipping fees. There were also some one-time registration fees paid to the Wisconsin Department of Natural Resources. The city of Milwaukee waived the purchase price in exchange for payment of these costs, so the \$25,000 that we paid essentially represents the cost of the lot. Not bad for a prime piece of urban real estate. Most of the other costs associated with the brownfield cleanup, estimated at \$50,000, actually related to the testing of the site before remediation could even begin. These costs were all covered by the city of Milwaukee using local governmental resources combined with federal grant funds.

Green Highlights

Following remediation, construction could finally proceed. In keeping with the schedule challenges endemic to the whole project, construction commenced in November at the start of Wisconsin's winter season. This resulted in a somewhat longer than hoped-for construction schedule, with the home finally reaching completion in August 2006. Here are some of the home's many green features.

Small footprint. The footprint of a building can be thought of in two ways: as the actual amount of space the building occupies on the ground, or as the amount of natural resources it consumes. Consequently, size is the most environmentally important decision in designing a home. Our home has a footprint of 900 ft². Local building codes forced the home to be larger than we felt we needed, coming in at 1,800 ft² total and 1,600 ft² of living space (200 ft² is lofted open space). The home is still significantly smaller than the average American home of 2,200 ft². By eliminating most hallways, it actually achieves a comparable living space. The home includes three bedrooms and two bathrooms.

Panelized construction. By constructing most of the home's basic framing in a factory, components could

be cut exactly to size, eliminating construction waste through more efficient production. Structural framing built in a factory also provides greater weather protection, minimizing damage by storing materials until just-in-time installation can be completed. The framing was made substantially from engineered wood, incorporating reused waste wood for stud materials as well as floor and roof joists and trusses. New wood used in the framing included wood from sustainably managed forests.



The Kaufmanns chose dual-flush Toto toilets for their home.

Geothermal heating and cooling. The decision to use a Geoexchange geothermal system in this green home was motivated by both practical and financial considerations.

Geoexchange systems provide heating and cooling at efficiencies that are far greater than those of most systems. They do so by using the nearly constant temperature of the earth's mass as a heat source in winter and a heat sink in summer. Our home uses a closed, horizontal loop of tubing placed at 6 feet below ground surface level and uses a water loop to transfer heat year-round through a 4-ton geoexchange system manufactured by Econar. A small amount of electricity is used to fire the system, and we buy renewable, wind-powered energy, provided through connection to the power grid, from the local electrical utility.

Compared to traditional heating and cooling, geothermal systems have

the lowest life cycle costs; they require minimal maintenance and they cut energy bills dramatically while increasing year-round comfort. Reductions can range from 50% to 75%, depending on the system and on energy use. Geothermal systems also cut greenhouse gas emissions. Up front cost is about twice the cost of a traditional system, but it can be recouped through annual savings early in the life of a home. In this case, we expect to recoup the full cost of the system in about five years. The fact that we had been obliged to remove 6 feet of soil across the entire site, as explained above, made this approach economically feasible. Also, the lot was slightly larger than the typical urban lot, which provided sufficient surface area for a horizontal loop system.

Indoor water use. Dual-flush toilets, long popular in Australia, Western Europe, and East Asia, offer two flush options: a standard flush for solid wastes and a lower-volume flush for liquid wastes and paper. The Toto Aquia residential toilets use 1.6 gallons at the full flush and 0.9 gallons at the low flush. A typical family of four will save approximately 7,000 gallons of water per year with this toilet, compared with a standard 1.6 gallon-per-flush toilet.

PEX plumbing. PEX offers many advantages over other plastic and metal tubing. It is healthier thanks to its completely nontoxic formulation, which is free of harmful lead, copper, and other minerals. It is less expensive because it is more durable; there are no joints behind the walls, and it is guaranteed not to corrode or erode. PEX is quiet, virtually maintenance-free, and affordable.

Reclaimed and recycled materials. Tons of construction waste is landfilled every day in cities across the country. By simply talking with contractors on the job, we were able to identify maple flooring being discarded from a warehouse redevelopment underway in an adjacent neighborhood. We reclaimed this flooring and had it refinished; it has found a new life in our home. Additional "waste" materials include reclaimed timber wood for the breakfast bar in the kitchen; recycled plastic decking material; recycled tile

made from industrial waste from the manufacture of luxury tiles; and recycled denim insulation used in the interior wall cavities to improve energy efficiency and dampen sound.

Sustainable and nontoxic materials. Floor surfaces not covered with reclaimed materials are covered with cork flooring, a renewable resource with insulating and sound-dampening

Star standards with a superior U-value as rated by the National Fenestration Rating Council. Clerestory windows also enhance natural lighting. LED lighting provides an added light source in the house. LED lights are substantially more efficient even than CFLs; they use approximately 5 watts per bulb as compared to a 75W incandescent bulb and they last nearly 25 years. Energy Star

steel serves as the roofing material on the home. Besides lasting at least 4 times as long as an asphalt roof, steel releases its heat faster in the summer (reducing cooling costs), and it allows for rainwater harvesting without the granules and petroleum that wash off asphalt. Because downspouts empty into rain barrels and an integrated garden irrigation system, no storm water will leave the lot. To maximize water efficiency, the landscaping will consist of native plants and will utilize xeriscaping principles.

In this home's urban setting, the water catchment system is designed primarily as a strategy to reduce rainwater runoff. The city of Milwaukee utilizes a combined sewer and storm water management system. Regrettably, most of the buildings in the city, including most homes, have drainage and downspouts connected directly to this system. This means that rainwater is captured and channeled directly into the system. Ultimately, and particularly during peak storms, overflow from this system discharges directly into waterways, including Lake Michigan. The city of Milwaukee frequently discharges untreated sewage into the lake as part of this overflow management.



The modern backside of this home is a unique element given its historic neighborhood location.

qualities. Bio-based soy insulation is also used in all exterior wall cavities. This product provides superior insulation by eliminating air infiltration; it is made from rapidly renewable soy and reduces overall energy loads. We expect to achieve R-values of 22 for walls and 42 for ceilings, while eliminating nearly all infiltration by using soy-based spray foam insulation. Nontoxic paints, adhesives, and sealants (low- or no-VOCs) are used throughout the house.

Integrated energy and atmosphere system. In addition to the geo-exchange system already discussed, the home uses ceiling fans combined with bathroom and kitchen vents to improve indoor air quality; the fans also provide cost-effective cooling. Eagle brand low-e windows are technologically designed to maximize heat gain in winter and sun reflection in summer, meeting Energy

appliances are also installed throughout the house to minimize energy use.

The decision not to incorporate a solar PV system in this green home was motivated by both practical and financial considerations. PV systems rely on sufficient solar access. This home site has a limited southern exposure due to the position of adjacent buildings and existing mature trees. In addition, the setbacks and siting of the structure, as required by municipal code, make it impossible to maximize the existing solar access. In short, solar access is insufficient for a PV system. We believe in making green homes accessible and affordable to mainstream home buyers. And our own budget would not have allowed for a PV system, even if it had been possible to install one.

Rainwater collection system. Recyclable, standing-seam Galvalume

Keeping on Budget

An overall project objective was to create a competitively priced, newly constructed green home in Milwaukee. Despite all the hurdles that we had to jump, the completed home achieved costs in the midrange of Milwaukee's new construction market, at less than \$200/ft². We wanted to demonstrate that mainstream green homes are no more expensive than traditional homes. By pragmatically blending cost savings methods with cost premium technologies, we were able to prove this point.

The up front cost of a geothermal exchange system, while twice the cost of a traditional HVAC system, will be paid back within five years. Other cost premium approaches included the use of no-VOC finishes, durable exterior brick cladding, and LED lighting. These premiums were offset through other cost-saving green measures. These in-

cluded the use of panelized construction, which reduces waste and makes for faster, more affordable construction; the use of reclaimed wood flooring, which was free except for the cost to move and refinish; a smaller footprint and efficient design to reduce resources needed and lower costs; and the use of tax rebates and government incentives where applicable—although they were minimally applicable in this home.

In many cases, green choices, such as the use of Energy Star appliances, were cost-neutral decisions. While our home is within the midrange of other nongreen homes on the market, keeping it cost competitive in up front pricing, we expect that it will consistently and significantly beat the market in overall long-term operational costs.

Since the home was just recently completed, actual utility costs and energy efficiency performance have not yet been measured. The September combined electric and gas bills, while not yet stabilized for typical use, were under \$100, suggesting that energy savings will meet expectations.

Taking a Pragmatic Approach

As we learned from planning our green home, an adaptable, pragmatic approach was essential to getting our project built within budget and in compliance with building codes. We made many compromises that married our idealism with the practical realities. The result is that the design pragmatically integrates green and traditional building technologies and systems and creates a practical, affordable, and sustainable end result.

One example of project pragmatism was a midcourse change in plan from a single structure to a project that includes a main home and a coach house.

Because the site is situated in a local historic district, many materials and design techniques were dictated by the local historic preservation commission, resulting in higher expenses. To offset



PEX tubing was chosen for its durability, lower cost, and nontoxic formulation.

these expenses, a coach house was added to provide a source of rental income. This income was used to help us meet our goal of keeping construction costs under \$200/ft². The rental income will help us to make the higher mortgage payments in the short term, and will provide our family with greater economic stability in the long term.

We hope that by living as an example, and by telling our story, we can make sustainability concepts accessible to more people in Milwaukee and beyond. To date, response to the home has been overwhelmingly positive. We have been so motivated by the experience of building this house that we now aim to transform the construction industry by modeling how beautiful and affordable green can be.

As part of this effort, I, together with my two partners founded a new company in early 2006. Pragmatic Construction was founded specifically as a green construction company—a company committed to a triple bottom line: people, planet, and profit. My partners and I plan to meet this commitment by building new homes and remodeling

existing homes in southeastern Wisconsin, using the principles of sustainable development. When today's construction techniques were developed, natural resources were inexpensive, and they were thought to be inexhaustible. Rising utility rates and construction costs have taught us that this is not the case. Clearly, there are environmental and financial incentives to modify outdated building methods. Despite this, the construction industry has been slow to change. Frustrated, my partners and I decided to take matters into our own hands. Pragmatic Construction offers the solution by focusing exclusively on, and delivering, beautiful home design; green, healthy, construction; energy efficiency; and lower construction costs. We aim to demystify green and make it accessible.



Juli Kaufmann is a principal in the green construction company Pragmatic Construction, LLC, operating in southeastern Wisconsin.

For more information:

For more on Pragmatic Construction, go to www.pragmaticconstruction.com.

For more on brownfield remediation, go to www.dnr.state.wi.us/org/aw/rr/rbrownfields.

For more information on the geoechange system used in the Milwaukee house, go to www.geoexchange.org.

For ore on green roofs, go to www.greenroofs.org.

And for information about green building in Wisconsin, go to www.wgba.org and www.greenbuihome.org.